Room Number _____

Seat Number _____

Student Number |__|__|__|__|__|__|__|__|__|

ANONYMOUSLY MARKED

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THE UNIVERSITY OF SYDNEY

CONFIDENTIAL EXAM PAPER

This paper is not to be removed from the exam venue

Mathematics and Statistics

EXAMINATION

Semester 1 - Main, 2020

MATH2921 Vector Calculus and Differential Eqs (Adv) (MATH2921 Vector Calculus and Differential Eqs (Adv))

EXAM WRITING TIME: 2 hours

READING TIME: 10 minutes

EXAM CONDITIONS:
This is a RESTRICTED OPEN book examination - specified materials permitted

During reading time - writing is not permitted at all

INSTRUCTIONS TO STUDENTS: TO DO – PUT SOMETHING APPRORIATE HERE – PERHAPS THERE SHOULD ALSO BE A STAT DEC

Please tick the box to confirm that your examination paper is complete. ☐

For Examiner Use Only

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1. Solve the following initial value problem:

$$u_{tt} - 4u_{xx} = 0 \qquad \begin{array}{ll} u(x,0) = \sin x & x \in \mathbb{R} \\ u_t(x,0) = \cos x & t \in \mathbb{R} \end{array}$$

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2. Calculate the flux of the vector field $\mathbb{F} = \begin{pmatrix} 3x^2y \\ -y^2 + xy \end{pmatrix}$ across the parallelogram with vertices at $(0,0), (1,2), (1,4)$, and $(0,2)$ in that order, oriented anticlockwise.

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3. Find all real critical points of the function

$$f(x,y) = x^2y^2 - 8x^2y + 15x^2 - xy^2 + 8xy - 15x$$

and classify them as local maxima, minima or saddle points.

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4. (a) Solve the following initial-boundary value problem:

$$\begin{aligned} u_t &= u_{xx} \quad x \in (0,\pi) \quad t > 0 \\ u_x(0,t) &= u(\pi,t) = 0 \quad t > 0 \\ u(x,t) &= \cos\left(\frac{3}{2}x\right) + 11\cos\left(\frac{11}{2}x\right) \quad x \in [0,\pi]. \end{aligned}$$

(b) What is the (exact) value of $u\left(\frac{\pi}{2}, 4\right)$?

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5. Find the solution to the initial value problem:

$$\mathbf{y}' = \begin{pmatrix} -2 & 0 & 2 \\ 0 & -1 & 0 \\ 2 & 0 & -2 \end{pmatrix} \mathbf{y} + \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

with

$$\mathbf{y}(0) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

What is the value of the solution at $x = 2$?

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6. Consider the function

$$f(x) := \begin{cases} x^2 + \dfrac{\pi}{2} & x > 0 \\ x^2 - \dfrac{\pi}{2} & x < 0 \end{cases}$$

(a) Compute the Fourier series of $f(x)$ with period 2π.

(b) Find an exact expression for the following series

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k^2}$$

(Hint: use your answer from part (a)).

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7. A thin plate is bounded by the function $y = -4x(x-1)$ and the x-axis on the interval $0 \leq x \leq 1$. Find the mass of the plate if the density at each point is $\rho(x, y) = 1 - y^2$.

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8. Write the general solution to the following ODE

$$y''' - 3y'' = 2\sin x - \cos x.$$

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9. Let $\mathbb{F} = \begin{pmatrix} P(x,y) \\ Q(x,y) \end{pmatrix}$ be a smooth planar vector field.

(a) Write $\nabla \cdot \mathbb{F}$ in polar coordinates.

(b) Let $u(x, y) := x^2 - y^2$ and $v(x, y) := y/x$ write $\nabla \times \mathbb{F}$ in terms of u and v.

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10. Consider the function defined for $x \in [0, 3\pi]$

$$f(x) = \begin{cases} 0 & 0 \leq x \leq \pi \\ \pi - x & \pi \leq x \leq 3\pi/2 \\ x - 2\pi & 3\pi/2x \leq 2\pi \\ 0 & 2\pi \leq x \leq 3\pi \end{cases}$$

(a) Sketch the odd periodic extension of $f(x)$ with period 6π.

(b) Which of the following Mathematica inputs would produce the coefficients of the odd periodic extension of $f(x)$ with period 6π?

(i)
```
1/(3*Pi) FullSimplify[Integrate[f[x]*Sin[k x],
{x, -3*Pi, 3*Pi}], Assumptions -> Element{k,PositiveIntegers}]
```

(ii)
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2/(3*Pi) FullSimplify[Integrate[f[x]*Sin[k x/3],
{x, 0, 3*Pi}], Assumptions -> Element{k,PositiveIntegers}]
```

(iii)
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2/(3*Pi) FullSimplify[Integrate[f[x]*Cos[k x],
{x, -3*Pi, 3*Pi}], Assumptions -> Element{k,PositiveIntegers}]
```

(iv) None of the above

(c) Suppose I considered the initial value problem

$$\begin{aligned} u_t = 8u_{xx} \quad & x \in (0, 3\pi) \quad t > 0 \\ u_x(0,t) = u_x(3\pi, 0) = 0 \quad & t > 0 \\ u(x,0) = f(x) \quad & x \in [0, 3\pi]. \end{aligned}$$

Would the formal series solution be a true solution? Why or why not? Justify your answer.

(d) Let $u(x,t)$ be the solution to the initial boundary value problem in part (c). What is the limit of $u(x,t)$ as $t \to \infty$?

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11. **Bonus:** Do not attempt this question until you have finished answering all the other questions on this exam. Tell me a joke.

This is the end of the examination paper